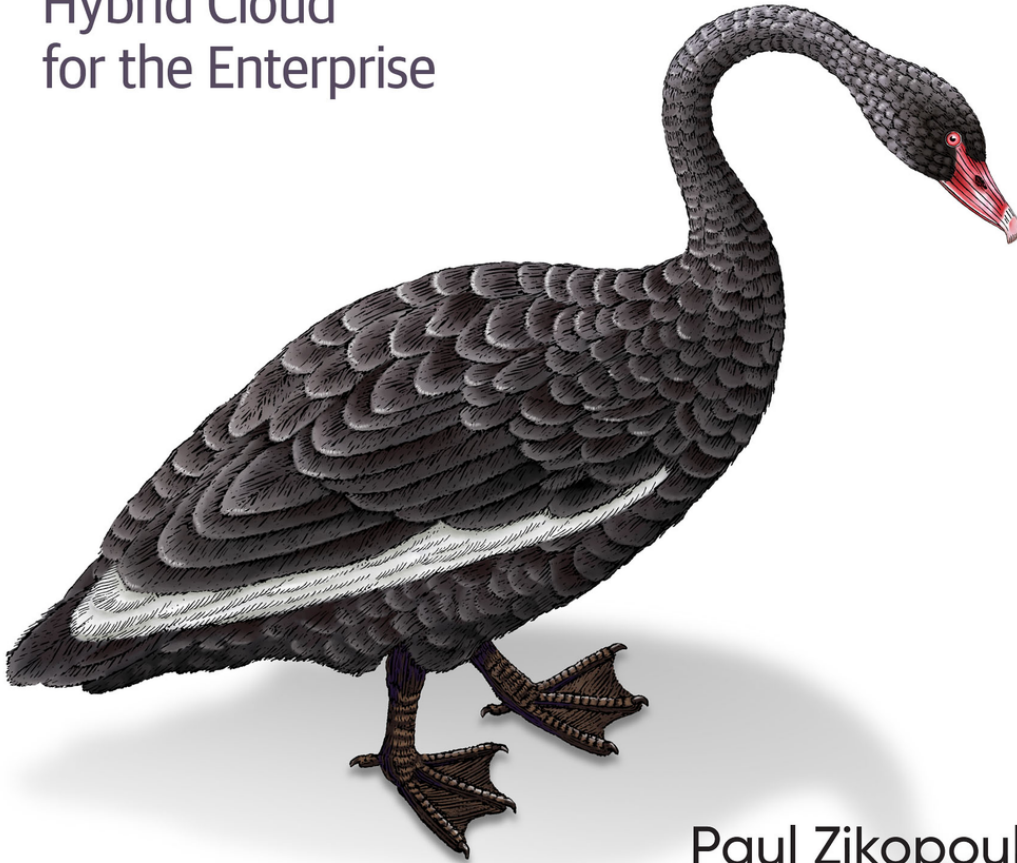


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Cloud Without Compromise

Hybrid Cloud
for the Enterprise



Paul Zikopoulos &
Christopher Bienko
with Chris Backer, Chris Konarski
& Sai Vennam

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Printed in the United States of America.

Published by O'Reilly Media, Inc., 1005 Gravenstein Highway North, Sebastopol, CA 95472.

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Cover Designer: Karen Montgomery

Illustrator: Kate Dullea

August 2021: First Edition

Revision History for the First Edition

- 2021-07-29: First Release

See <http://oreilly.com/catalog/errata.csp?isbn=9781098103736> for release details.

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978-1-098-10373-6

[LSI]

Preface: Who This Book Is For

The five of us got together and wrote a book that might (at times) be too technical for business leaders, but “too business-y” for technical people. If this is a complaint you’ve got after reading this book...perfect! While we definitely put some technical stuff in here, if you’re looking for a book on how to use `kubectl`—the CLI interface for Kubernetes—or use `generator` to create a `ConfigMap`, this is not the book for you.

Our experience has shown us—more than any one of us wants to admit—that too many business people don’t truly understand the technology they’re signing the bills for (or for that matter, that technology’s potential). Nothing good comes of this: late projects, missed expectations, excessive costs, and worse yet—lost opportunities. If business teams lack the “chops” to challenge IT or understand technology discussions within their organizations, it’s akin to using a decades-old word translation algorithm. For example, one famous algorithm took a US phrase, “The spirit is willing but the flesh is weak,” and translated it into Russian as, “The vodka is good but the meat is rotten.” We think you get our point. Business people—if your spirit is willing, we can’t promise you good vodka; but we can promise you a good (in fact, a great) foundation to a cloud strategy that lets you lead from the front and helps you be the leader you want to be.

At the same time, this book is for all those propeller heads (we know you well, we are you) who get enthralled with the underpinnings but fail to realize that technology for technology’s sake doesn’t help a business succeed. If you fit that profile, this book is going to really help drive your career because there are not a lot of people that have your technical backgrounds. Beyond helping the organizations

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you work for succeed, this book will give you options to “pond jump” into myriad leadership roles across your company and accelerate your career. For whatever pothole or mountain of accomplishments our authoring team has, we can tell you that it all came from a deep technology start with layers of business experience and acumen added over the years. We’re not saying be like us (that may or may not be a good thing), but we are telling you this: technology people who understand business outperform technology people who don’t understand business.

So just whom is this book for? We think anyone. You might be informed on all aspects of cloud, or you might just be getting started and have little understanding of how it works, why it works, or how to implement and *scale* a successful cloud strategy. While we targeted this book for all business leaders, no matter who you are, we think this book is going to help you. Naturally it’ll help those leaders who are not sure where to start, but we also think our personal experiences will greatly benefit those who have started and are unsure of where to go next. With all that said, we’re confident there is a cohort of business and technical individuals that this book is really going to resonate with. These folks typically work in large organizations that have been around for a while. These organizations have a mix of generational technology. They have monolithic apps they don’t want to touch because those rock-solid apps literally keep “the lights on,” but they’ve also built some cool mobile stuff using service-oriented architecture (SOA) or microservices (don’t worry if you don’t know what those are, we get into them in **Chapter 4**). If you work big-box retail, oil and gas, banking, airlines, government, insurance, consumer packaged goods (the list goes on and on), we can pretty much *guarantee* you that you fit into this group. This book is for you because you’ve definitely got some cloud going on, but you may not be making the progress you’d hoped for or (quite commonly) you’re not getting the returns you thought you’d be getting and wondering if you’re doing it right.

This book requires no previous knowledge of cloud. The topics we introduce will be helpful for a vast audience, ranging from business leaders to technical folks across all industries. If you’re a business leader, you’ll know just the questions to ask, the corners to look around, and the ideas to challenge as you tactfully plan your forward-thinking cloud strategy. If you’re a technical person, you’ll understand how your expertise can drive forward the organizations you work for with a solid understanding of how cloud challenges (especially for large-sized companies) are much different than a startup’s “Swipe your credit card and let’s get going” approach.

Whoever you are, we think the concepts you need to know to successfully scale a cloud strategy and get all the value out of cloud can be found in the pages that follow.

Conventions Used in This Book

The following typographical conventions are used in this book:

Italic

Indicates new terms, URLs, email addresses, filenames, and file extensions.

Constant width

Used for program listings, as well as within paragraphs to refer to program elements such as variable or function names, databases, data types, environment variables, statements, and keywords.

Constant width bold

Shows commands or other text that should be typed literally by the user.

Constant width italic

Shows text that should be replaced with user-supplied values or by values determined by context.

TIP

This element signifies a tip or suggestion.

NOTE

This element signifies a general note.

WARNING

This element indicates a warning or caution.

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NOTE

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Our Collective Thank Yous and Acknowledgments

This book would not have been possible without the insights and efforts from some really fine people we know. Terry Bird and Meg Murphy jumped in on the idea and never let go of it from the get go, across divisional line changes and reorganizations; they are always so forward thinking and we thank them for that. Thank yous are also in order for Shelia Beresznewicz (who handled all the contracts and lawyers), Paul Amrbaz and Pamela Chace (our web presence peeps), Bill “Mr. Quantum” Minor (helped review some sections), and Filipe Miranda (for his insights on SELinux and access controls—a true technologist with a deep understanding of everything cloud). Of course, a collective thank you to Linda “Eagle Eye” Snow (and not because she’s from Philly) for always being willing to read a chapter and use her sharp eyes to catch almost anything wrong.

Finally, we want to heartfully thank (although at times we cursed their deadlines) the O'Reilly crew's personal efforts that went into getting this book from an idea we had while Slacking each other wondering how long on average it takes for people to realize they are double-muted on a WebEx call to what's in your hands today. Some people show up at their job every day to earn a paycheck, but we're glad these folks go to work to build something: Michele Cronin, Mike Loukides, Sharon Cordesse, Kristen Brown, and Rachel Roumeliotis...thanks!

Our Personal Dedications and Reflections

Paul Zikopoulos

If you've read any dedication for any book I've written in the last five years, you've concluded I'm a liar. Seriously, you can't keep telling people you're writing your last book and keep writing, so I won't make any promises not to put pen to paper again. That said, I know one thing I will never do again: write a book with three people named Chris, especially when two of them share the same first letter of their surname! No offense, these are some of the most talented guys I know—but it sure is confusing and makes calls difficult because you can't even say, "Chris B..." (add to this Chris Backer is pronounced like Baker). On a serious note—to a fantastic group of professionals and friends, some new to me, some not, and one like a brother.

Let me tell you, it's a lot of work to write a book. Lots of cursing (at times, my kid thought I was watching an HBO show in my office) aimed at the people that were there from the get go (Terry Bird, one of the finest people I know, was an f-bomb victim) across weekends and long weeknights wondering if you're making a difference. After all, it's not like I'm writing *50 Shades of Hybrid Cloud*, so these works don't bring me early retirement. Every book I write, I learn. And with that I write to tell stories (truthful ones), to wrap technology in a manner that you can understand and with hopes that you might find a smile here and there as you read through it. So why do I write? Honestly, my parents instilled me with a never-ending thirst for learning; I'm all too well aware that one of the biggest blessings in my life right now is a Caller ID that says, "Mom or Dad."

To Brad Arnott. The most talented guy that refuses to see it. I mean who else decides to golf and breaks 80 in 3 years, decides to be a fisherman and goes pro, decides to be a marksman and wins awards, decides to do anything and becomes great. Take a moment Brad, look in the mirror, and know the awe you inspire.

To my wife (Kelly) and kid (Chloë)—forever by my side—nothing more needs to be said.

Professionally, to the group of people that inspire me on the daily to try to be a better person and use my brains for something bigger. Stephanie Trunzo, Deb Jenson, Deb Bubb, and Kelly Lyndgaard—you're the kind of ladies any father wants their daughter to follow—you've become sisters, and I stand in awe of you on the daily. To Chris Hugill and Bob Picciano, who continue to inspire me and leave me in awe of their knowledge—I wonder if these guys even use Google. Within IBM, there are so many people to mention, so I'll just random sample the list. I took over a new division in 2021—lots of new team members that I asked to buy into a new team ethos, a mission (with a steep climb) and to all of them that followed, thank you. To Drew Valentine and John Teltsch who have always been a text message away. And to a new boss, Sebastian Krause, who has just the right amount of everything to be tremendous—there's a quality of leadership that drives you where you need to be driven (even if I don't like it at times) and gives you the autonomy to create...that's his recipe. A shout-out to Steve Astorino (an incredible leader) and to the transformation he has done with culture—your ideas are even bigger than the engine in that supercar you drive, and the Toronto lab adores you for it. Finally, it

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seems I can't write a book without thanking, consulting, or coauthoring with Rob Thomas. For 10 years, you've been 20% pushing me (in a way I don't like) into personal growth zones with your challenges and 80% inspiring me to never take my foot off the knowledge gas pedal. What you're doing at IBM is something beyond special.

Remember everyone, health is wealth—so stay positive and *forever* test negative!

A handwritten signature in black ink that reads "Paul Zikopoulos" with a stylized flourish at the end.

Christopher Bienko

To those friends and family that inspire me to live authentically, thank you. It would be an understatement to call 2020 merely a challenging year—I think, for all of us to an extent, it has been a rare opportunity to evaluate what's important to our lives and make an honest measure of how we are spending the precious time we have. And most remarkably of all, it was (and continues to be) a strangely universal experience shared by every person across each corner of the globe. Those events are few and far between.

It's my fervent hope that collectively we hold on to the silver linings of a taxing year: the remarkable quiet and calm of those first weeks of lockdown, the added time at home with family or friends in lieu of globe-trotting or commuting, the recognition of the selfless work carried out day after day by others to make our lives safe and secure. I've been an avid outdoorsman for the whole of my life, and it's been wonderful to see how COVID-19 has rekindled in the public mind a renewed draw towards natural spaces, public lands, and healthy ecosystems. With greater public interest comes additional pressure on these delicate environments, but likewise a mutual interest in their stewardship and conservation. My belief is that the more stakeholders we have in protecting public lands and natural systems, the more likely we are to succeed in preserving these treasures for the future. Let's keep that momentum going.

In hindsight, it seems halfway absurd to have started writing a book amidst all the uncertainty and personal reckonings of the last year. None of this, frankly, would have been possible without the unwavering commitment and talent of my coauthor and contributors. Paul, Chris B., Chris K., Sai, and our incredible publisher—you have my sincere thanks and gratitude.

A handwritten signature in black ink that reads "Chris Bienko" with a stylized flourish at the end.

Chris Backer

As an early professional, I spent countless hours reading O'Reilly technology books, learning and growing my skills. With every new project that I embarked on, I often sought out an O'Reilly book on the topic. I am truly grateful to have had this opportunity as a coauthor for a topic that has become one of the most transformational technology shifts of my time. To my coauthors, it has been an absolute pleasure working with and learning from each of you. A special thank you to Paul Z for the invitation to contribute to this book. You are an inspiration to many!

To Anna Shugol, Elton de Souza, Filipe Miranda, Patrick Fruth, Roberto Calderon, and Vic Cross (aka The Z Acceleration Team). You've led our brand into a new era of Hybrid Cloud—and in the midst of the most challenging event our society has faced in modern times. You all are the “best of the best.” I am truly grateful for your dedication and entrepreneurial mindset. #TogetherWeWin

To my wife (Anna), thank you for your support and encouragement over the years. You push me to be a better person every day. And to my incredible boys (Ty, Colton, and Caden): each of you make me so proud! Set your goals high and never stop learning.



Chris Konarski

When I look back at all my professional and personal accomplishments, it all started with an inner hunger and work ethic that was instilled in me by my parents John and Grace. The lessons I learned about hard work, competition, setting high goals, and how to treat people have served as my recipe for success. I dedicate this book to you and thank you for always pushing me outside my comfort zone to be “even better.”

To my boys, Ryan and Noah, set high goals and go after your dreams. If it is hard then you are growing; learn to be comfortable with being uncomfortable because that is where the magic happens. Feed your inner curiosity and fire with goals you never thought possible. I will always be there to love, support, and push you to greatness wherever your passions take you.



Sai Vennam

A voice, written or spoken, is the single most important thing a person has—it enables one to communicate their thoughts, expand others' beliefs, and influence the world. I'd like to thank my colleagues and mentors that helped me create my own voice—Adam Gunther, Jason Gartner, Andrew Hoyt, Jason Goode, and Nathan Smith.

Personally, I'd like to thank my older brother, Ram Vennam, who always served as a role model to me from a young age all the way through blazing the trail for me as a developer and leader at IBM. Finally, I'd like to thank my wife Reena, my sister-in-law Belinda, and my parents for always being there to support me—the only reason I've made it this far in my career is the love and care from those dear to me.



Introduction

There are three things that dominate 90% of technology conversations today: AI, cybersecurity, and cloud—we're guessing you figured out what this book is about by the cover. (And we know you're wondering: the animal is a black swan—quite fitting when you think about the state of the world when we started writing the book.)

Collectively our authoring team has well over a century of IT expertise and that means we've seen a lot of stuff (we had a different word in here, but it looks like the editor changed that). If you've ever stumbled across the **Gartner Hype Cycle**, you'll know that just because a technology is the "talk of the town," it doesn't guarantee success in the grand scheme of things. We've found that even for those technologies that find their way out of Gartner's "Peak of Inflated Expectations" and into the "Plateau of Productivity" part of their Hype Cycle curve, the calculus can still become a damp squib (a wet firework that fails to go off). Time and time again, we see an upstart technology that becomes all the rage, but just doesn't quite manage to stick the landing the first time: Knowledge Graphs (we still think something will happen here), Learning Gamification (at least the way most people implement it), governance (most organizations have a least effort to comply approach), and Hadoop...just to name a few.

Let's be clear on something: *Cloud is not in a hype cycle!* In fact, it's one of the most significant platform inflection points of our time (especially when you start thinking about cloud in the way we describe it in this book). It's true, sometimes we look at each other and say, "I can't even keep a cell phone connection, how is the world going to totally run on cloud?" Make no mistake about it, cloud is here and it's here to stay—you need to pivot the way you think about cloud if you want to get all you can out of it.

However, we've found that most businesses aren't getting the value out of the cloud that they were expecting (the mental model around cloud was too narrowed by the hype). Some might feel that there's more value yet to be uncovered, other organizations are guilty of not even knowing what to expect (just like AI, many jumped on the notion that the gains will be instant—like magic), and some are even repatriating public cloud workloads back to their traditional runtimes. Why? Too many compromises.

How do you get to cloud without compromise? It's not a vendor thing. We're not here to tell you the only way to get cloud without compromise is to migrate everything to a particular vendor's public cloud. For example, while we think there are nuances for specific types of applications that make the IBM Public Cloud a standout (highly regulated industries like finance), other public cloud providers have their own characteristics that are differentiated too (take a look at the catalog and configuration options on AWS).

There are two things to really grasp before you can get to a cloud without compromise. First, internalize what will become a mantra of this book: *cloud is a capability, not a destination*. Cloud without compromise means you're shifting your mindset from a place to an *operational model*. Second, cloud without compromise means you're embracing a unified distributed hybrid cloud strategy. A *hybrid cloud* unifies public cloud, private cloud, and on-premises infrastructure with consistent management and orchestration to create a single, flexible, cost-optimal IT infrastructure (the distributed part means it doesn't matter what vendor you select for your public cloud resources. Let the gravitational pull to a cloud vendor be the strengths—its cloud capabilities—that uniquely serve your business. This is why we'll advocate again and again for looking at cloud as a capability (instead of a destination), so that your business and the things it needs to be successful can run (and interoperate)

anywhere to everywhere. The outcome? Companies with a no compromise cloud strategy are empowered businesses that can:

- Combine best-of-breed cloud services and functionality from multiple cloud computing vendors
- Choose the optimal cloud computing environment for each workload
- Move workloads freely between public and private cloud as circumstances change

When no compromises are made, companies are free to pursue their technical and business objectives more effectively and cost-efficiently than they would otherwise be able to through any single public or private cloud vendor alone. In fact, according to one recent [study](#), companies derive up to 2.5x more value from a hybrid cloud strategy than from following a single-cloud, single-vendor approach.

Initially, hybrid cloud architectures focused on the mechanics of transforming portions of a company's on-premises datacenter into private cloud infrastructure, and then connecting that infrastructure to public cloud environments hosted off-premises by a public cloud provider. Today's hybrid cloud architecture needs to be focused less on physical connectivity, and more toward supporting workload portability across all cloud environments (the location or vendor doesn't matter) and on automating the deployment of those workloads to cloud environments (again, agnostic to vendor or location) with the most gravitational pull for your application's needs.

Several trends are driving this shift. As part of the next critical step in their digital transformations, organizations are building new applications (and modernizing legacy applications) to leverage cloud native technologies—technologies that enable consistent and reliable development, deployment, management, and performance across cloud environments and across cloud vendors, including on-premises infrastructure. Specifically, they're building and transforming applications to use microservices architectures, which deconstruct unwieldy monolithic applications into smaller, loosely coupled, reusable components focused on specific business functions. And they're deploying these applications in containers—lightweight executable units that contain only the application code and just enough of the virtualized operating system dependencies required to run it. These technologies serve as a foundation for enabling businesses to drive a new-age culture of productivity, such as development cycles that last days or weeks (instead of quarters or years) and the inclusion of highly effective methodologies like test-driven development, continuous integration and continuous delivery (CI/CD), A/B testing, and more.

At a higher level, public and private clouds are no longer physical “locations” to connect together. For example, many cloud vendors now offer public cloud services that run in their customers' on-premises datacenters. Private clouds, once run exclusively on-premises, are now often hosted in off-premises datacenters, on virtual private networks (VPNs) or virtual private clouds (VPCs), or on dedicated infrastructure rented from third-party providers (who may happen to be public cloud providers).

What's more, infrastructure virtualization with the aid of automation (infrastructure as code) allows these environments to be created, on demand, using any resources located behind (or beyond) a firewall. This takes on added importance with the advent of edge computing, which offers opportunities to improve global application performance by moving workloads closer to where data is created and consumed.

Cloud native development makes it possible for developers to transform monolithic applications into units of business-focused functionality that can be run anywhere and reused within a variety of applications. A standard operating system (like Linux) lets developers build any hardware dependency into a container. And Kubernetes orchestration and automation delivers granular, set-it-and-forget-it control over container configuration and deployment—including security, load balancing, scalability, and more—across multiple cloud environments. (Don't worry if you don't know anything about what we just wrote; this book is going to teach you about all of this.)

We talk a lot in this book about the untapped cloud value that remains for those downtrodden on their cloud journeys. To the uninitiated, they might think, “Yes, that means if I move everything to a public cloud vendor, I will get all this value.” That's not what it means—remember, cloud is a capability, not a destination. But the right cloud strategy (which you'll learn how to craft in this book) will certainly deliver:

Improved developer productivity

This helps expand the adoption of Agile and DevOps methodologies, and enables dev(elopment) teams to build once and deploy anywhere.

Greater infrastructure efficiency

With on-demand granular control over rapidly provisioned compute and storage resources, development, and IT operations, teams can optimize their spend across public cloud services, private clouds, and cloud vendors. Hybrid cloud also helps companies modernize applications faster and connect cloud services to data on cloud or on-premises infrastructure in ways that deliver new value.

Larger breadth and depth

Access to a larger number of services spanning AI (which includes machine learning and deep learning) capabilities, storage, data processing, analytics, automation, and more. These are services that cloud vendors have tailored for your business's needs with the expertise and depth to run them at enterprise scale.

Improved regulatory compliance and security

A unified platform lets organizations draw on best-of-breed cloud security and regulatory compliance technologies, implementing Zero Trust security and compliance across all environments in a consistent way.

Overall business acceleration

This includes shorter product development cycles; accelerated innovation and time-to-market; faster response to customer feedback; faster delivery of applications closer to the client (like edge ecommerce); and faster integration with partners or third parties to deliver new products and services.

If you want cloud without compromise, without a doubt in our minds, almost every medium to large organization requires a hybrid cloud strategy that starts with capabilities and not destinations (let that fall into place afterward) and folks...that's what this book is all about.

Chapter 1. Cloudy Skies Are the Best Forecast Ever

A cherished bottle of Rémy Martin Louis XIII Cognac is blended from 1,200 different cognacs aged up to 100 years, with each bottle representing the career achievements of generational cellar masters. Parmigiano-Reggiano Stravecchione's (the Italians refer to it as the "King of Cheese") complex flavor and texture are a result of no less than four years of aging. Bruce Springsteen spent an unbelievable six months editing the lyrics of his famous song "Born to Run."

Oh, the sung praises of slow mastery—great things take time. But while cognac, cheese, and songs can be *slowly* aged into masterpieces, technology years are like dog years: things change, and change happens quickly. While the romanticism of slow change is valuable for some things, today's businesses and institutions need more speed to market, more flexibility, and nimbleness to respond to changes in the economy, buyer behavior, supply chains, geopolitical realities, climate change, and more. If you had to prioritize a list of all the things you could do to deliver a better product or service in the most optimized fashion, infusing technology into all parts of your business should be at the top of the list.

The changes we've seen in the past few years are rewriting the basic behaviors and assumptions we've had about IT. Technology has the very real effect of redefinition. Businesses must rethink how they create and deliver value, how they compete, how they transact, and ultimately how the business itself works. Technology provides a strong basis for business and institutional innovation by creating a lingua franca for data, applications, and workflows where an ecosystem of ideas from customers, suppliers, and partners can be brought together.

In the US, the Black Friday shopping bonanza¹ is a great example of how technology changes IT practices. Many retailers still impose a blackout period against pushing any technology changes during those dates because they don't want to risk anything that could impact profitability in the final days of the year. Then microservices came along, which allowed a few retailers on the leading edge to change those behaviors. For example, Amazon's website isn't a monolithic page of logic; it's an eloquently

orchestrated set of microservices that come together and work harmoniously to form a real application. Jeff Bezos, former Amazon CEO, once famously said, “Build services or look for a job somewhere else.” This is exactly why we tell clients their goal isn’t to build applications, but rather to compose them! Amazon’s web pages are hundreds (if not thousands) of small services that have single jobs. Building software this way completely redefined how that organization creates and delivers value, how it works, how it competes, and how it transacts.

With technology, almost everything about a business—its logic, purpose, and differentiation—can be rendered in code, making digital innovation the most powerful way to drive transformation and change (or to pivot during a crisis). The emergence of public clouds has, until recently, led the debate on one of the fastest ways to prepare a business for the future; however, a closer examination suggests that most workloads (about 80%) have yet to move. Most companies have only taken on lifting and shifting what they could “as is” to the cloud, or building “greenfield” (net new or born on the cloud) apps on public clouds.

As it turns out, the mission-critical workloads that run large businesses and institutions often span extensive IT estates that include traditional on-premises investments and multiple clouds—private, public, and even on edge devices. Another inhibitor to the realization of cloud value is that almost every client we’ve ever talked to has invested with multiple cloud vendors and technologies. Add to this business operations that span geographic locations, each with unique government and regulatory requirements, and you’ve got a jumble of experiential and implemented cloud solutions without interoperability. The end result? The inadvertent creation of a hybrid cloud without a strategic approach, leading to messy, high-friction barriers to innovation and realization of your cloud aspirations. Our recommendation? Embrace the diversity by building a purposeful hybrid cloud strategy (we’ll tell you how throughout this book). If you’re familiar with Hadoop, its fate was sealed because too many business leaders thought they could build a single analytics strategy around it. The cloud is the same: a single cloud strategy just isn’t realistic thinking.

Thrivers, Divers, and New Arrivers

Without question, the events of 2020 were an awakening for many. In 2021, the effects of the COVID-19 fallout continue to impact nations the world over, changing commerce and lifestyles in ways that would have seemed unthinkable just a short while ago. The only certain fact in these uncertain times is that “things” will never be the same, even with a vaccine. Whatever the future will be dubbed—“back to normal,” “the new normal,” or our preference “the new abnormal”—leaders must reckon with a new set of challenges that span the gamut from societal reform, to global health, to new ways of conducting business and education. One area of upheaval that has often been treated as an afterthought but has been profoundly changed by the events of these last few years: the way business is (and will be) done.

The COVID-19 pandemic has been an eye opener on many fronts. From a business point of view, most companies have come to realize (COVID aside) they are simply not modernized (prepared) for a digital

economy. Each of us has had experiences that could fuel stories to be told. We've heard early anecdotes of people attempting to order toilet paper online from a big-box store, then walking by a toilet paper endcap in a brick-and-mortar store because they didn't want to panic buy, only to find out that their online order could not be fulfilled and was cancelled!

Another big-box store had an outage and could not process transactions; yet another familiar household name left customers in a web chat queue for hours, waiting for answers to basic questions such as order status or finding open locations. Some people waited for hours on both the phone and in chat queues to see which would resolve the problem (or get a response) faster, and discovered that these two channels made the business seem like different companies—so much for your omnichannel support (insert your experience here...). There's no coincidence that all of these big-name stores weren't born online. They (and us as customers) figured out quickly that their transformation journeys weren't nearly on the well-trodden path that they thought they were. The truth is that they realized that their hard-won “digital transformation” wasn't anywhere near as effective as they thought.

One thing is certain: large or small, those companies that didn't have a true pulse on their digital transformation before most certainly do now. They've seen customer satisfaction drop to nonexistent levels because of a lack of digitized self-service resolution and overall order fulfillment frustrations (can you say, “negative Net Promoter Score”—don't snicker, it's a real possible score). For many, it took months to set up curbside pickup; many vendors' online SKU catalogs *still* don't match what's in-store; search engines “found” merchandise that was out-of-stock, but didn't tell the customer until checkout (just when you thought you finally found what you needed); and many recommendation engines surfaced highly sought-after items in stock-out positions to go along with your in-stock purchases...frustrating.

These are all examples of a value chain that ignored user centricity and collapsed upon itself—supply chain disruption that was carelessly surfaced to clients in need at the most inopportune time. These companies were all guilty of not knowing what they already could have known—an enterprise amnesia if you will. It's perfectly fine to be sold out of a hot item; it's quite another to take a user right through to checkout, and then surface an out-of-stock position or direct customers to a brick-and-mortar store based on inventory data that's more stale than the “Last Day for Sale” rack at the seediest donut shop you've even been in. But was this avoidable? Yes!

These actions (or perhaps better said, inactions) cost sales, eroded customer loyalty, diluted trust, and opened the front doors with a “Welcome to our Home—Take What you Want” mat for modernized businesses to steal market share—and so some of them did. COVID-19 has caused many businesses large and small to fail; you just need to read the headlines to see what happened. While it's nearly impossible to understate the devastating impact of COVID-19 on global commerce and people's well-being, there is something few are speaking about: the harsh toll inflicted on businesses who've been talking (for a long time) about things like resilience and agility but didn't make them true priorities (some might call it “walk the talk”). Contrast this with those who prioritized these values and were well prepared to seize unexpected opportunities for growth. These companies became digitally touchless

and fully (and consistent) omni-channel; it didn't matter if you were on a mobile app or website, phone or chatbot: your interactions had the same capabilities in either medium. That isn't to say there wasn't a downside or pain for even the most prepared companies; however, it's fair to note that if most companies had a six-month warning of what was coming, they would have done a lot of things differently and prepared in different ways. That's the point.

During the unprecedented challenges of COVID-19, many companies were divers, while some were thrivers. Consider two famous branded US-based craft and bulk food stores. Both enabled their omni-channel presence with backend business operations to support a seamless curbside pickup experience within days of closing their retail stores (the thrivers). In contrast, because of outdated monolithic ordering systems and static infrastructure, a number of large grocery stores and retailers took weeks or months to get their order fulfillment processes in line with the current reality—despite having far more IT resources and budget. Others found their way as new providers as they rose to the occasion: the new arrivers. For example, think about the growth of internet-connected fitness class apps like DownDog (provides a studio-like yoga experience in the comfort of your home) or fitness equipment companies like Ergatta (rowing) or Tonal (pulley weight training with a large screen assistant), and so on. These new arrivers differentiate from those that thrived (like Peloton). As you can see, across all industry, some thrived, some arrived, but many dived.

In another instance, a sporting goods store couldn't process orders (or they took minutes to process) leading to a record number of abandoned online shopping baskets. Desperate customers ran to what were once virtually unknown vendors who could swiftly answer questions or tell them their order was processed in a timely manner (and where an order confirmed meant you were actually getting the item, which wasn't a guarantee with the divers)—a warm welcome to the new arrivers!

There are far more bad stories than good—many more divers than thrivers or arrivers—and the consequences of shaken consumer confidence continues to outlive the pandemic. The one undeniable advantage shared by arrivers and thrivers alike was a well-thought-out renovation and innovation framework. As we move into emerging-from-pandemic discussions with clients, it's clear that their digital modernization plans just got accelerated by five years as digital-first becomes even more critical to their future.

Business Vaccination: The Arriver's Guide

As the world deploys COVID-19 relief in the form of vaccination, a business “vaccination” has long already existed—it's the activation of a well-proven innovation and renovation strategy (we'll give you a framework for understanding renovation and innovation at the end of **Chapter 2**). That's a bold statement, and the point isn't to make light of the seriousness of COVID-19 from a personal health perspective, but to seriously consider the components of an agenda that has the net effect of moving the mindset (and the business for that matter) away from “survive” and into “thrive and arrive.”

Figure 1-1 shows a well-trodden and proven path to renovate and innovate your IT estates. We'll discuss each of the steps in the sections that follow.



The Arriver's Guide

AI

Automation, optimization, prediction

Modernization

Flexibility, built for the cloud

More security

Hackers, crackers, spies, oh my!

Performance demands

More digital transactions

More resiliency

Uptime is business time

Cost Takeout

Undoubtedly, COVID-19 has hit many bottom lines and that will surely mean (to many) reduced budgets and sensitivities to investments that don't have direct impact on profit. One cost-savings strategy is cloud computing. As it turns out, many who think they'll just flip their apps to a public cloud often come up short on cost savings (some spend more—read **“The Cost of Cloud, a Trillion Dollar Paradox”**) or never fully realize the cloud's benefits; this is why experts agree that about 80% of applications haven't become “cloud-ified” (we'll talk about this later in the book). Folks that come up short are often tying the word cloud to a destination and not a capability.

An additional cost-takeout winner will be consolidation, which can impact hardware and software costs (never underestimate the ability for good hardware to reduce software core licenses). The use of cloud native technologies like containers and Kubernetes are huge, as consolidation and virtualization play into the hands of any cost-savings strategy. This all enables business in an economically savvy and concentrated fashion. This also frees resources (computational and financial) that can be funneled toward other critical areas that make up an infrastructure agenda—such as resiliency, security, and performance. Bottom line: cost-savings opportunities will come with other considerations that need to be well understood and planned, and those that think the cost savings opportunity is strategically but a simple click away will be as successful as those business leaders who think AI is magic.

Resiliency

This matters more than ever before: in the new “abnormal,” consumer behaviors of buying, selling, and socializing have been drastically altered. Even after populations have been vaccinated en masse, long-term buying habits, underwriting car insurance in a hybrid workplace environment (this hybrid is a destination), banking (and more) will be permanently changed. Consider the effects of a potential future COVID-19 flare up: the storefront you offer clients for transactions has to keep on running; your contactless payment method must work (in some places you weren't even allowed to take cash for purchases). These solutions need to be in place before the next potential challenge arises. In other words, getting caught on your heels again will not be an option. You should ensure you're familiar with key performance indicators (KPIs), such as mean time to repair (MTTR) and mean time between failure (MTBF), recovery point objectives (RPOs), and recovery time objectives (RTOs), among others. In plain speak, these KPIs provide answers to questions like: “How long can you typically go without an outage?”, “How often will an outage be expected as a function of time?”, “How much data can you stand losing?”, and “How fast do you need to get back up and running?” You'll need to know the answers to these questions like the back of your hand.

Performance

The shift to more online procurement and contracting means more performance demands on the infrastructure. Think about panic buying. While much fun can be made of toilet paper hoarders amidst lockdowns and supply chain breakdowns, think about critical drug therapies and interventions. For many, medical prescriptions became a legitimate panic and so began a race to pharmacies to fill them. Pharmaceutical systems had to keep running, even under the unprecedented transaction levels they experienced in the first weeks of the COVID-19 pandemic (not just from individuals buying their medicine, but from pharmacists needing to pull medical records, allergy information, cross-drug interactions, warnings, and more). There were pharmacies during this challenge that didn't miss a beat, and those that sadly got beat up. It's also worth a moment of thought to consider how the accelerated movement to contactless payment methods helped to mitigate the spread of disease. Those payment systems not only needed to work (resiliency) but they needed to respond quickly (performance)—before the COVID-19 crisis began, if point-of-sale systems went down, accepting cash only was merely a frustration. As COVID-19 infections became more and more prevalent, these systems going down literally meant “no sale.”

Security

As more and more businesses harden their online presence in response to growing customer demands, security is paramount. Evidence suggests that hackers used the COVID pandemic as an opportunity to up their attack frequencies—**Forbes** noted a 238% rise in attacks on banks and a 600% increase on cloud servers between January 2020 and April 2020! We expect to see concepts such as Zero Trust, Security Information and Event Management (SIEM), and Trusted Execution Environments (TEEs) move from “great to talk about” to real plans to do it. The more and more people transact online, the more risk we face with fraud and identity theft. The more data that's in-flight or at-rest, the more it needs to be protected (hint: today, the world isn't doing the job well enough). In fact, contactless pay providers like Apple Pay, PayPal, or Android Pay (among others) should see a surge because credit card information doesn't have to land or flow through the vendor, thereby lowering the risk of credit card details falling into the wrong hands. What about the protection of data at rest? We're betting (because we were shocked at how much we didn't know until we knew) most don't fully appreciate how much of a role even storage considerations play in a truly hardened cyber-resilient strategy.

Now consider the fact that the amount of data breaches and data heists for crypto ransoms has never been more plentiful (2021 is proving this in monthly intervals) and dangerous (a disproportionate number of data ransom schemes are mounting up against healthcare facilities—and now think about COVID).

Fact: corporate boards everywhere are asking all kinds of security questions and the job market is thriving (last we checked, we saw well over 200,000 Information Security jobs). This means that if your answer to the question “Where did the hacker go with our data?” begins cheekily with “He ransomware,” that won't be cute or funny to anyone. This suggests that companies (now more than

ever) need to align, protect, and manage threats to their ecosystem (both on-premises and off-premises)—and COVID-19 has made it readily apparent just who is (and who isn't) ready to offer those assurances to their customers.

Modernization

Modernization is a must-have mindset for both thrivers and arrivers. Flexibility will be a difference maker—and flexibility leads you to the cloud. The term *hybrid multicloud* (a mix of multiple cloud vendors and cloud technologies on-premises and in public cloud providers) is a hot topic these days because its very name embodies flexibility, and this is a space you've got to know and focus on. The emerging space of distributed cloud (the same cloud technology across multiple cloud destinations, delivered by technologies like IBM Cloud Satellite, Amazon Output, and Google Anthos) is becoming a hot topic as companies examine the management and skills costs associated with multiple cloud vendors. Whatever the hybrid cloud genre (multicloud, distributed, mix of on-prem and public, or all of them), dynamic (metered) pricing models for consumption of resources is today's marketplace trend. That's a cloud capability. Infrastructure that's flexible in capacity to scale up (or down) in a utility-like fashion at the most granular levels—down to the minute. That's a cloud capability. A service catalog to self-provision modern development environments or applications to accelerate workflow. That's a cloud capability. We could go on (and we will).

But there are right and wrong ways to go about building a cloud strategy. To do it right, we've developed a mantra for this book: "Cloud is a capability, not a destination." What does that mean? If you think of the cloud as a destination, you'll be thinking about AWS, or Azure, or Google, or IBM. And you'll be tempted to unify behind one of them. That's not how the cloud era is playing out for individuals or companies. Think about your personal computing infrastructure. If you have an Apple phone, tablet, or laptop, you're probably using iCloud for something (like backing up your contacts). If you watch movies on Netflix, you're using Netflix's cloud providers (yes, they run on two different clouds that must work together seamlessly to deliver you new episodes of *Grey's Anatomy*). Same for Spotify. And if you use Gmail, you're in Google's cloud.

Businesses are no different. A company's cloud journey usually starts with a few pilot projects in different departments, often because no one wants to wait for IT to approve new hardware: perhaps it's easier to train your "playing around" AI model using Google Collab's free GPU resources versus going to the "Department of No!" (Finance) to get some GPUs. By the time the C-suite starts thinking about a "cloud strategy," the company is already on multiple clouds. It's already multicloud. (This is classic "Shadow IT": you see it in choice of databases, cloud platforms, and the lot.)

But multicloud itself is a dead end if there's no effort to provide a uniform programming and operational interface to those clouds. You shouldn't have to care about where your data is located or where your software runs. That's why the "cloud" isn't a destination. If it is, you'll end up with a fight between marketing doing BI on AWS, finance doing business planning on IBM, and R&D doing AI on GCP. These efforts all need to collaborate. Hybrid clouds are all about integrating multiple cloud

providers and in some cases, creating a single unified interface to them (distributed cloud)—and, for that matter, to your own on-premises data and applications. That’s where the value of the cloud lies: in the flexibility of not having to worry about where your data is, or where your code is running. That’s what it means to say, “Cloud is a capability.”

According to McKinsey (and many others) 80% of enterprises are choosing to work in a hybrid multicloud environment, now and in the future. Tying these value statements to a destination does nothing but cheat the enterprise.

NOTE

Most cloud strategies have been focused on building new applications on the cloud (they are called *greenfield* apps) or a lift-and-shift of existing applications to the public cloud. We refer to this cloud strategy as “Cloud Chapter 1” (an epoch in the journey to cloud, not an actual chapter in this book). We think the hybrid cloud—and its ability to capture the true value of its capabilities—will be the next chapter in the cloud storyline and thus we often refer to it as “Cloud Chapter 2” (the next epoch of the cloud journey). Think of it as getting deeper into the cloud story, where the characters, the plot, and immersion thickens and provides more value.

If you’re having a conversation about modernization, bringing the right apps and services up to par with containers, orchestration, and microservices-based architectures should be among your highest priorities. You should definitely be investigating Kubernetes (K8s). K8s platforms such as Red Hat OpenShift Container Platform (OCP) are becoming the new “operating system.” Open Container Initiative (OCI)—compliant storage containers, enterprise-ready Kubernetes orchestration of those containers, and automation via Ansible and Terraform are the essential pillars for creating an agile IT foundation. (Don’t worry if those words don’t mean anything to you yet; we’ll dig into them throughout this book, but understand them as the technologies that make “Chapter 2” of the cloud possible.) If you build containerized applications to the open standard OCI governance structure, those containerized applications can be provisioned anywhere (on-premises or in the public cloud), regardless of the cloud vendor. Modernizing applications and services on containers deployed with K8s orchestration is one of the best strategies that you can invest in. It doesn’t matter whether your applications are solely apps born on the cloud (people refer to these as cloud native; they were built with modern approaches like microservices and containerized from the start), legacy apps that run your business, or a mishmash of them all (which is more likely the case). This is a winning strategy.

It’s been said that basketball is a game where a single player can take a team to a title; however, in soccer, it’s your weakest link that can keep you from winning anything. With that in mind, remember that the orchestration platform (K8s) always emulates the ethos of the infrastructure. Was the infrastructure built for resiliency and fast recovery? Was it built to scale? Was it built securely right down to the virtualization level and for Confidential Computing? This all matters; these are all good questions to ask. The modernization journey is a soccer game, not a basketball game.

AI

Can there be a tech-minded book written without the word AI these days? We think not. Yes, it's hyped, and the world tends to overhype what's possible in the next two years. But we can promise you that it often underestimates the profound changes brought about by the technologies that stick around for the next five to ten years. We think that in the future "AI" won't stand for Artificial Intelligence, but rather Ambient Intelligence. Why the word "ambient"? Think of ambient lighting; when done correctly, you don't really notice that it's there, but it's doing its job. AI will serve a similar role in our day-to-day lives and jobs: it will be everywhere, and become just a natural part of our environment. We will stop thinking of it as strange and different.

To us, the opportunities of AI can be summed up as: automation, optimization, and prediction. AI will drive costs down and client engagement up. Without question, successful businesses will unlock the power of data with AI—according to **Forbes**, up to 73% of data goes unused in most organizations—that's why your AI needs an IA (information architecture). What's more, automation will be unavoidable; some reports suggest that enterprises can use automation to reclaim 120+ billion hours per year spent on low-value work. As companies move from defining their AI progress via algorithm counts in the dozens (at best) to the thousands, discussions will move well beyond how fast and accurate you can train your AI. We will be talking about inferencing speeds (scoring), lifecycle management (the moment you publish your AI algorithm is the moment it's out of date), sourcing data, explainability, bias, and more. Despite being a cloud book, we're going to give you a skills hotlist that any successful enterprise adopting AI needs to master: language, automation, and trust. Cloud is going to be the conduit for great AI...so these themes go hand in hand.

So Why Are Cloudy Skies the Best Forecast Ever?

From golf birdies to bankers, free drops to rain drops, highways to back roads, cloud technology affects them all (and all parts in between). Hybrid cloud is the next major shift in the evolution of IT...think about it, when any new IT architecture emerges and spreads, it has the potential to change the world and the concept of hybrid cloud (which will eventually just be called "cloud") will completely change the way renovators renovate and innovators innovate.

We believe we're at a key inflection point on the continuum of human history and innovation. Our world is set to pivot from a world of automation where humans are supported by technology to a data-rich ambient intelligent world where the technology is supported by humans. To us, this pending shift means that the largest wave of business transformation architecture is just ahead, and it'll be driven by three major shifts:

1. The modernization of critical workloads to build scalable applications at speed.
2. The adoption of Kubernetes to orchestrate those applications across any environment, no matter the vendor or location of the compute runtime.

3. This will all compound to serve as a springboard for the growing use of operational AI and edge applications to create data-driven insights that shape business outcomes.

As you go through this book, we think it'll be obvious that only a hybrid cloud architecture can provide a consistent, standards-based approach to development, security, and the operational hygiene that'll be required. It is a smarter open architecture that allows for workload portability, orchestration, and management across multiple environments. We won't be able to stress it enough in this book: get into the cloud as capability mindset (as opposed to destination) and you'll be a leader of an organization that simply does better.

¹ The nickname for the Friday following US Thanksgiving Day where stores offer deep discounts for the Christmas shopping rush; many refer to this event and the days from it up until Christmas as “make or break” in terms of sales for the current year.

Chapter 2. Evolution of Cloud

In the first chapter, we introduced you to the thrivers, divers, and new arrivers; we boldly declared the mantra of this book (“Cloud is a capability, not a destination.”), and gave you a guide to digital transformation. In this chapter, we'll discuss the evolution of cloud, where it's going, and some things to keep in mind as you begin (or continue) your capability journey.

Are You on the Intranet, Internet, or Extranet? Nah—Just Internet

To get a feeling for what we're talking about, consider the ubiquitous phrase: the internet. At one time, many would subdivide internet technologies into an access control list of sorts. There was talk of the EXTRANet: a controlled private network limited to vendors, partners, suppliers, or other authorized groups. Today, most would say an extranet provides access from outside the firewall to resources inside the firewall for a value chain based on membership. For example, when you return equipment to a store that sends your broken item back to the manufacturer who deals with the warranty claim, the store is likely to log in to the manufacturer's site and open a ticket, but that's about it. Even so, bringing your items to the store is a necessary step because the store has access that the customer doesn't.

At work we used the intRAnet: an internet that was behind a firewall and dedicated to providing information and services within an organization, usually to the exclusion of access to outsiders. In other words, if you didn't work at the company, you had no access to it. This intranet would have department operational manuals, performance reviews, sensitive documents, code repos: all the stuff used to run your business that is nobody else's business.

Then it started. A rallying cry around a set number of standards by groups like the Internet Engineering Task Force (IETF) or the World Wide Web Consortium (W3C). Communities ratified all kinds of standards, from documents (HTML), to fetching those documents (HTTP), to the concept of time, and all parts in between; soon enough, things just became referred to as the INTERNet. Think about it, when was the last time you used a word other than internet to describe these kinds of networks? When was the last time you heard someone at work say, “I found it on our intranet?” Answer: you were probably dancing the Macarena and weren’t embarrassed doing it. (Sure, there will be some roles where this differentiation matters, but this isn’t how business talks and it isn’t how most people think.) The bottom line: intranet, extranet, and internet just became internet.

Are You on a Private Cloud, Public Cloud, or Community Cloud? Nah—Just Cloud

History has a way of repeating itself. Think “cloud.” Cloud-talk today centers around terms like public clouds, community clouds, and private clouds; but work has started on a bunch of standards that have started to evolve and unify the patterns.

We think cloud will follow the same naming path as the internet and will settle on hybrid cloud (again, cloud as a capability not a destination), or just cloud for short. We’re already seeing technologies that support cloud computing in general (just like those that supported internet technologies). Think of the standardized open source technologies (many of which we cover in this book) such as Jenkins, Terraform, Ansible, Docker, and Kubernetes (K8s), all supported by major vendors for the orchestration and management of cloud native applications. These are the technologies that are creating integrated hybrid clouds.

With this in mind, we want to officially anoint our book’s mantra (you’ll hear it time and time again through the book); say it with us: “*Cloud is a capability, not a destination!*” If we can agree that cloud is a capability, then it’s perfectly aligned to where IT wants (and needs) to go: decentralized, flexible, performant, open, and secure.

Just like the “prevent defense” (an American football defense that’s happy to let the offense move downfield but holds defenders further back to try to prevent a major score) we think prevents a win, a “one location and vendor cloud fits all” approach won’t work either. Believe us, we’ve seen it fail, we’ve even seen some overzealous competitor sales (and IBM) reps try to push it. It often doesn’t work and it’s not the right tone for your business. Companies need the flexibility to run their workloads across any platform without having to rewrite everything as they go.

History Repeats Itself: From Granularity of Terms to General Terms

This is why our book’s mantra, “*Cloud is a capability, not a destination,*” encompasses a mindset shift you (and the organizations you work for) have to make to fully exploit the opportunities that those

servers in the cloud offer. We've seen success in the cloud—mostly companies producing greenfield apps; that's great if you're a startup or building net new in a roll your own (RYO) fashion. But what about established investments? Many of these companies looked at the cloud as a pure cost-saving opportunity, and while some had success (most in limited fashions that didn't nearly match the promises), others were left unpleasantly surprised. While the cloud can offer you some cost savings (cloud toe dippers typically start with archive or test), it all depends on the running workloads, data gravity, and your business. But cloud can't just be about cost savings—in fact, over time it'll prove to be more about agility than cost savings. Sure, that capability has potential to reduce the obvious costs, but it's about putting your business in the best position possible to deliver.

But here is the cold hard truth: we estimate that a mere **20% of the potential value** from cloud computing is being realized today by mature enterprises via the public cloud. As you will learn about in this book, cloud is more than cost savings; it's a renovation opportunity that delivers flexibility. When you think about it the right way, you'll start to think beyond “lifting and shifting” as-is applications to the cloud. For example, releasing software frequently to users is usually a time-consuming and painful process. Continuous integration and continuous delivery (CI/CD) can help organizations become more agile by automating and streamlining the steps involved in going from an idea, a change in the market, a challenge (like COVID-19), or a business requirement to ever-increasing value delivered to the customer.

Adoption of cloud being a capability and not a destination opens up your world to rethinking how applications are built, delivering continuous streams of improvement to engage your user base (think about how often apps get updated on your iPhone), and so much more. We believe this mantra will have you look at the technologies that support the cloud (Kubernetes, Docker, Jenkins, Terraform, and oh so many more) and appreciate how they make for agile organizations and encompass the entire end-to-end lifecycle of an idea, pivot, or change of course you need for your business.

So how do you access the 80% of value that's yet to be unlocked? You embrace a hybrid strategy and all its components (on-premises, public cloud, multicloud, distributed cloud). In fact, don't just embrace it—demand it. When vendors come knocking, talk about capability. Discuss the ability to dynamically access and move capacity that your business needs regardless of location. Not just on-premises or in a public cloud—but taking an app from one cloud provider and seamlessly running it on another; remember, we want the ultimate flexibility here. Start to think about breaking down the monoliths and containerizing those apps with a fundamental change to the operating model.

Building a hybrid cloud strategy and playbook is essential if you want to unleash the full potential of your cloud; in fact, **studies** suggest such a strategy can deliver 2.5x more value than a public-only strategy (the 20% of current captured cloud value we alluded to earlier).

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